2024 International Joint Conference on Neural Networks (IJCNN) | 979-8-3503-5931-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/IJCNN60899.2024.10649905

Manufacturing Domain QA with Integrated Term Enhanced RAG

Yijun Bei[a,*], Zhibin Fang[a], Shenyu Mao[a], Shuyi Yu[a], Yan Jiang[a], Yining Tong[b] and Weimin Cai[b]
[a]School of Software Technology, Zhejiang University, Ningbo, China
{beiyj, zhibin.fang, shengyu, yushuyi, yan.jiang}@zju.edu.cn
[b]Zhejiang Jiyun Education Technology Co., Ltd, Ningbo, China
{tongyn, cwm}@jobgroup.cn

***Abstract*—Large Language Models (LLMs) have demonstrated powerful capabilities, yet LLMs face issues like hallucination in certain domain-specific areas. Consequently, an increasing number of domain-specific models are emerging. The current paradigm for domain-specific models involves training with domain data, followed by the employment of Retrieval-Augmented Generation (RAG) to mitigate hallucination issues. However, in precision-critical domains such as manufacturing, if the knowledge documents are of low quality or contain noise, the context retrieved through simple semantic matching by RAG may not necessarily benefit model output. Additionally, there can be issues like getting "lost in the middle" due to irrelevant or excessive context. To overcome this, we introduce the *Integrated Term Enhancement Methodology* (ITEM). Inspired by Chinese educational methods focused on key term elucidation, ITEM extracts and explains critical terms precisely from knowledge documents to form a comprehensive Term Dictionary for retrieving terms and explanations to enhance query capabilities. This methodology refines query responses by providing more accurate and contextually relevant information. To assess ITEM's effectiveness, we utilize the Chinese Mould Manufacturing Dataset (CMMD) and Contextualized Adaptive Response Assessment (CARA) metric method. Our experiment demonstrates that ITEM significantly outperforms existing retrieval enhancement Dense Retrievers by over 17.0% in accuracy while requiring only 80% of their token length. Moreover, the accuracy of our method exceeded that of GPT-4 by 5.0%. This advancement represents a significant leap in context-specific retrieval in LLMs, especially beneficial for specialized domains. The results underscore ITEM's potential as a transformative method in the field, offering new perspectives on integrating domain-specific knowledge into LLMs.**

I. Introduction

The rapidly evolving field of natural language processing (NLP) has witnessed significant strides with the advent of Large Language Models (LLMs) such as GPT-3.5 [1] and GPT-4 [2]. These models have demonstrated their versatility and robustness across various applications. However, their effectiveness in specialized domains is often hampered by challenges such as model hallucination and the lack of in-depth domain-specific knowledge [3], [4], [33]. This deficiency has spurred the development of domain-specific LLMs, meticulously designed to cater to the intricate needs of specific fields such as healthcare, biomedicine, and oceanography [5]–[8]. Such models are increasingly recognized as vital tools in professional contexts, highlighting the ongoing demand for more sophisticated, specialized LLMs.

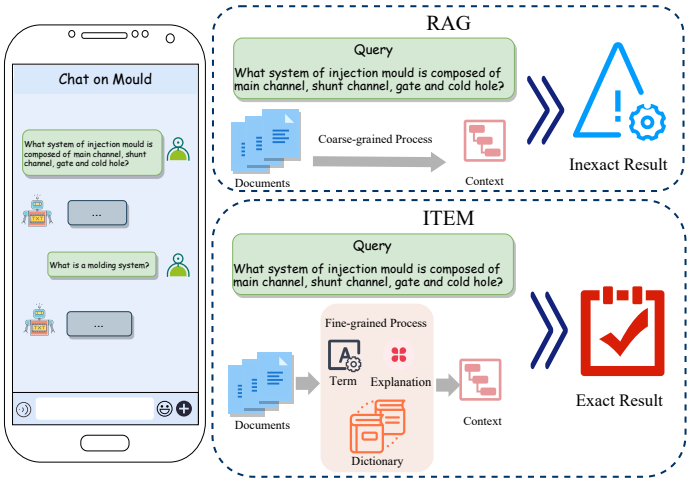


Fig. 1. The comparison between the RAG and our ITEM method. Our ITEM method provides fine-grained context, whereas RAG can only provide coarse-grained context.

To address these challenges, implementing these models typically involves fine-tuning them with domain-specific data, followed by using Retrieval-Augmented Generation (RAG) technology to fetch highly relevant knowledge from knowledge documents as context. This approach enables large models to reference pertinent knowledge, thereby providing more reasonable answers and reducing hallucination issues [12], [24]. However, in the manufacturing domain, especially in question-answering scenarios within manufacturing, where the accuracy and reliability of information are crucial [9], [10], RAG has certain limitations. First, it requires high-quality domain knowledge documentation, but compliant documentation is rare in manufacturing. Moreover, the context retrieved solely based on similarity may not always benefit model generation. In addition, in the practical application of RAG, it has been found that there is a phenomenon of "lost in the middle". When the context is very long, the model will ignore the useful knowledge in the middle part of the context and only pay attention to the beginning and end [35].

Our novel *Integrated Term Enhancement Methodology* (ITEM) seeks to extend their application to the manufacturing

*Corresponding Author.

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 02:14:36 UTC from IEEE Xplore. Restrictions apply.

sector, a domain still untapped in fully harnessing domain-specific LLMs' capabilities. We adopt a pedagogical strategy from Chinese vocational education that emphasizes the clarification of key terms for specialized subjects. This approach entails the distillation of textbook terminology into a Term Dictionary. In application, terms and their explanations are integrated with practical exercises, reinforcing domain-specific vocabulary and enhancing field expertise. Drawing from this, ITEM augments LLMs using a domain-specific Term Dictionary derived from knowledge documents, thus enhancing response quality and specificity in the mould manufacturing domain. Fig. 1 shows the difference between our method and the RAG method.

Furthermore, we introduce the Chinese Mould Manufacturing Dataset (CMMD), a curated collection of industry-specific texts and Q&A pairs to evaluate ITEM. Additionally, this dataset represents the first collection of industry-specific NLP data in the manufacturing sector, contributing significantly to future testing of various models and methods in this domain.

The Contextualized Adaptive Response Assessment (CARA) is a multi-choice question evaluation method introduced for assessing language models in natural language processing, especially in domain-specific contexts. Through rigorous experimentation, we demonstrate ITEM's superior performance in enhancing the accuracy and relevancy of LLMs in the mould domain. The experimentation indicates that ITEM, maintaining a consistent token length, surpasses the leading RAG method by an accuracy margin exceeding 17.0%. Additionally, to attain accuracy on par with RAG, ITEM necessitates merely 80% of the token count that the latter employs. The experiments showcase notable improvements over conventional LLM applications, validating ITEM's effectiveness in bridging the gap between LLMs and industry requirements.

The paper is structured to provide a comprehensive review of domain-specific LLMs and retrieval-augmented generation, followed by an in-depth exploration of the CMMD dataset and ITEM methodology. The conclusive section reflects on the implications of our findings and proposes directions for future research.

In summary, our contributions are as follows:

- We present the *Integrated Term Enhancement Methodology* (ITEM), a novel approach enhancing Large Language Models in the mould manufacturing domain by using a domain-specific Term Dictionary inspired by key term elucidation strategies in Chinese education.
- We introduce the Chinese Mould Manufacturing Dataset (CMMD), the first industry-specific NLP dataset in manufacturing, aimed at evaluating ITEM's effectiveness in domain-specific contexts.
- We introduce the Contextualized Adaptive Response Assessment (CARA),a method to evaluate LLM's accuracy in multi-choice question.
- We demonstrated ITEM's effectiveness through rigorously extensive experiments, showcasing significant improvements over existing retrieval methods by over 17% in accuracy and even exceeding GPT-4's performance by 4%, showcasing its potential in specialized domains.

II. Related work

A. Domain Large Language Models

These models, such as OpenAI's GPT, Google's PaLM, and Meta AI's OPT, have demonstrated versatility and robustness in a wide range of applications [1], [2], [15], [16], [25]. However, the performance of these models is limited in some areas. Hence, there is a trend to expand the capabilities of large language models (LLMs) to various domains [27], [28]. In healthcare, for example, MedGPT utilizes electronic medical records to predict future medical events, showcasing the potential of LLMs in predictive analytics [6]. Similarly, BioGPT, with its training on a vast corpus of biomedical literature, excels in interpreting complex biomedical terminologies, thus aiding in advanced research and diagnostics [7]. OceanGPT, another domain-specific model, has been instrumental in marine sciences, providing insights into oceanographic data and environmental studies [8]. These specialized applications of LLMs highlight their adaptability and the growing importance of tailored AI solutions in addressing specific industry challenges and advancing knowledge in diverse fields. Particularly, there are two mainstream implementations of domain-special models. One is based on a pre-trained model that uses a large amount of domain data for training, and the other is based on an already aligned model that uses retrieval to enhance the query with related context [12].

This research extends the utility of LLMs to the mould manufacturing sector, exploring their application in this specialized field. We aim to demonstrate how LLMs can be effectively adapted to the unique linguistic and technical challenges of the manufacturing domain, particularly in areas like mould design and production where precision and domain expertise are crucial [9], [10]. This exploration is part of a broader trend to tailor LLMs for specific industries, potentially revolutionizing how these sectors leverage LLMs to aid information processing and decision-making.

B. Retrieval-Augmented Generation

In question-answering tasks within specialized domains, applying Large Language Models (LLMs), or even domain-specific LLMs trained with domain data, directly to Q&A models still results in significant issues of model hallucination [3], [4]. To address the hallucination issues in domain-specific LLMs, the most prevalent method currently is Retrieval-Augmented Generation (RAG), leveraging external information sources such as databases, and knowledge documents to enhance the query [29], [30], [34]. RAG involves providing relevant documents as context to the language models, enriching the original input. There are mainly two methods for searching relevant documents: sparse retrievers and dense retrievers [17], [18]. Sparse retrievers utilize lexical matching techniques, relying on the occurrence and frequency of words (e.g., TF-IDF [19], BM25 [20]) to assess the relevance between queries and documents. On the other hand, dense

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 02:14:36 UTC from IEEE Xplore. Restrictions apply.

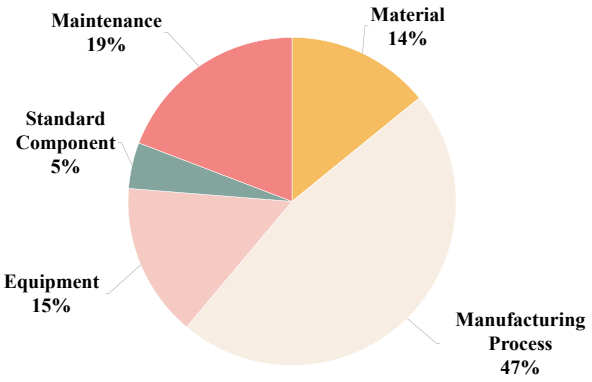


Fig. 2. Distribution of knowledge documents in the CMMD dataset across five primary topics within the mould manufacturing domain.

retrievers employ deep learning techniques, especially pre-trained neural networks like M3E [21], to generate vector representations of queries and documents.

However, sparse and dense retrievers tend to retrieve documents with high content similarity to the query, which may not always provide information that is genuinely useful for the query. The content found might be superficially similar to the query but not necessarily helpful for solving specific problems or tasks. Crucial insights often reside in documents that do not demonstrate immediate semantic salience but contain pertinent information [32]. Moreover, this approach has certain demands on the quality of external information sources. The presence of a substantial amount of irrelevant information will lead to a "lost in the middle" problem.

Therefore, this work aims to explore a new method to overcome the limitations of current retrieval-augmented generation and enhance the accuracy and reliability of models in domain-specific question-answering tasks.

III. Dataset

We have developed a dataset named Chinese Mould Manufacturing Dataset (CMMD), which includes data relevant to mould manufacturing such as manufacturing processes, material properties, etc. It is comprised of a meticulously curated collection of textual corpus and a robust assembly of question-and-answer pairs.

The textual corpus of CMMD is an extensive compilation encompassing 3,140 documents segregated into five definitive topics, equipment, standard components, manufacturing processes, and maintenance. Each document has been carefully selected from a variety of sources, including schools, such as course lesson plans and practice problems, and public Internet documents, such as essays and patents. This collection has been refined through a careful process, employing regular expressions for text processing and hash-based algorithms for deduplication, thereby mitigating the risk of model overfitting and eliminating redundancies and extraneous content. All the information has been verified by the mould-domain professionals. Fig. 2 illustrates the proportional distribution of the CMMD's knowledge documents across the five topics.

In addition to its textual corpus, CMMD encompasses a segment of 5,703 question-and-answer pairs, meticulously categorized into two subsets for model fine-tuning and validation. The fine-tuning set, comprising 5,203 instances, is constructed from specialized course exercises, ensuring a comprehensive training ground for the models. The validation set includes 500 carefully multi-choice question choice instances from recent mould industry qualification exams spanning the last five years. This deliberate selection process is designed to mirror real-world industry benchmarks, providing a stringent and authentic standard for evaluating the models' effectiveness in practical scenarios.

IV. Method

A. Task Formulation

The task of Retrieval-Augmented Generation for question answering is formally articulated as follows: Presented with a query q, a contextual subset C_{context} from a designated database, serving as external knowledge pertinent to q. This retrieved-context C_{context}, in conjunction with q, is then input into a Large Language Model (LLM) to generate an appropriate response.

B. Overviews

In this section, we introduce the ITEM, which is segmented into two principal stages: *Term Dictionary Construction* and *Term Explanation Enhancement*. During the first stage, given a set of documents $\mathcal{D} = \{D_1, D_2, \ldots, D_n\}$, we process each paragraph to extract terms and their corresponding explanations, thereby constructing a term dictionary $\mathcal{TD}$. For the second stage, when a query q is posed, we search within $\mathcal{TD}$ to identify the top-k most pertinent terms and explanations, as C_{context}. This C_{context}, in conjunction with q, is subsequently utilized to fine-tune the LLM and is also used in the inference.

C. Term Dictionary Construction

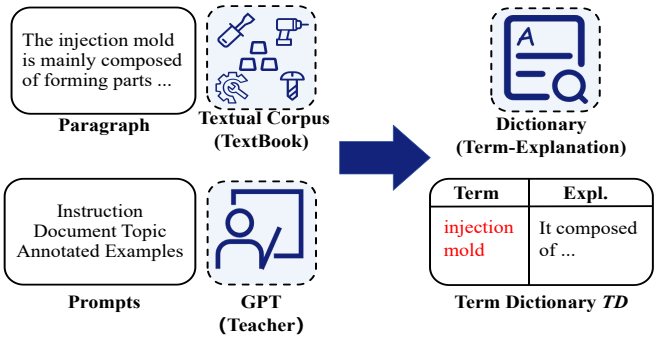


Fig. 3. Construction of the Term Dictionary $\mathcal{TD}$.

Figure 3 illustrates the process of Term Dictionary Construction. For all documents in the CMMD, we traverse each paragraph and employ GPT-3.5 to extract relevant terms and explanations. To ensure the specificity of the terms, we use

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 02:14:36 UTC from IEEE Xplore. Restrictions apply.

TABLE I
EXTRACT KEY TERM EXPLANATION FROM PARAGRAPH

Paragraph	Term & Explanation
The injection mold is mainly composed of forming parts, a pouring system, guiding parts, a launching mechanism, temperature regulating system, an exhaust system, and supporting parts, and sometimes lateral typing and core pulling mechanisms.	**Injection mould:** It is composed of forming parts, a pouring system, guiding parts, a launching mechanism, a temperature regulating system, an exhaust system, and supporting parts, sometimes including lateral parting and core pulling mechanisms.

the topic of each document and construct prompts that focus on extracting content related to its topic.

In-context learning (ICL) is a technique wherein the model uses a small set of annotated examples to generate data, enhancing its capacity for producing relevant and precise outputs [11]. We implement ICL by providing a representative set of labeled instances——extracted terms and explanations from paragraphs—and developing task-oriented instructions to augment the prompts, guiding GPT-3.5 to extract terms accurately.

To ensure the relevance and precision of the extracted terms and explanations, we employ the TF-IDF method to calculate their similarity to the original paragraphs. We adopt a threshold criterion, only terms and explanations with a similarity score above a predefined threshold of 0.8 are retained. This selective inclusion refines the construction of the final Term Dictionary, denoted by $\mathcal{TD}$. The construct of the $\mathcal{TD}$ is formulated as

$$\begin{aligned} \mathcal{TD} &= \bigcup_{i=1}^{n} \bigcup_{p_{ij} \in D_i} \{\text{TE}_{ij} : \text{Sim}_{\text{TF-IDF}}(p_{ij}, \mathcal{TE}_{ij}) > 0.8\} \\ \mathcal{TE}_{ij} &= \text{GPT-3.5}(\text{INS}, \text{TP}(D_i), \text{AE}, p_{ij}) \end{aligned} \tag{1}$$

In this formulation, INS specifies the instruction, $\text{TP}(D_i)$ denotes the topic of document D_i, and AE represents the annotated examples. The Term Dictionary, represented as $\mathcal{TD}$, is defined as $\mathcal{TD} = \{(t_k, \mathbf{E}_k)\}$, where t_k is the k-th term and $\mathbf{E}_k$ is the corresponding set of explanations, such that $\mathbf{E}_k = \{e_{k1}, e_{k2}, \ldots, e_{kt}\}$. Moreover, the low similarity is filtered to ensure that the model does not add additional content in the process of extraction and summary to ensure the quality of terms and explanations. This methodology streamlines the extraction of essential terms and their explanations from each paragraph in the domain-specific corpus. An illustrative extraction process from a paragraph to terms and explanations is depicted in Table I.

Finally, we employed the embedding model to convert all terms $\mathbf{T} = \{t_1, t_2, \ldots, t_k\}$ into embeddings. These generated embeddings $\mathbf{T}$ were then stored in the vector library, denoted by $\mathbf{V}$.

$$\mathbf{V} = \{v_1, v_2, \ldots, v_k\} \tag{2}$$

where v_k represents the embedding of term t_k, and is obtained by applying the embedding function Embed to each term, i.e., $v_k = \text{Embed}(t_k)$ for each $t_k \in \mathbf{T}$.

D. Term Explanation Enhancement

Due to the specialized structure of our Term Dictionary, our methodology encompasses a three-stage process to ensure precision and relevance in term retrieval. Figure 3 illustrates the process of three stages.

Retrieval of Top k Terms. The first stage is essential for discerning the most relevant terms relative to a query within the retrieval context. This stage involves harnessing the vector library $\mathbf{V}$ to match top-k terms that are most relevant to the query, which guarantees that only the most important terms advance to the next stage. The following expression encapsulates this stage's analytical framework:

$$\text{TopK}(q, \mathbf{V}) = \underset{t_i \in \mathbf{T}}{\arg \text{top-k}} \ \text{cos_sim}(q, v_i) \tag{3}$$

In this equation, $\text{TopK}(q, \mathbf{V})$ identifies the top k terms from the set $\mathbf{T}$ based on their relevance to the query q. This relevance is quantified using the cosine similarity measure $\text{cos_sim}(q, v_i)$, which evaluates the similarity between the query q and the vector representation v_i of each term in $\mathbf{V}$. The operation $\arg \text{top-k}$ is applied to select terms with the highest similarity scores, thereby determining the most pertinent terms with q.

Explanation Selection. This stage involves selecting the most contextually aligned explanation for each term t_i in $\text{TopK}(q, \mathbf{V})$ from the initial retrieval. It emphasizes the importance of context-specificity, ensuring that each term's explanation enhances the understanding of the original query:

$$\mathbf{RTE} = \bigcup_{i=1}^{k} \{(t_i, \text{Sim}_{\text{TF-IDF}}(q, E_i))\} \tag{4}$$

In this formulation, $\text{Sim}_{\text{TF-IDF}}(q, E_i))$ represents the function that determines the most similar explanation from the set E_k for the query q. For each term t_i in $\text{TopK}(q, \mathbf{V})$, the selected explanation is the one that has the highest similarity with q. The resulting set of Retrieved Terms Explanations ($\mathbf{RTE}$) comprises these term-explanation pairs, enhancing the interpretability and relevance of the original query within the specific domain context.

Concatenation of RTE with Query. During this phase, the set $\mathbf{RTE}$, comprising meticulously refined term-explanation pairs, is amalgamated with the original query q, thereby serving as the contextual component C_{context}. Such integration culminates in the formulation of an Enhanced Query ($\mathbf{EQ}$). This augmented query $\mathbf{EQ}$, imbued with contextual richness and specificity, is then utilized in the fine-tuning and evaluation stages of the learning process.

V. EXPERIMENT

A. Metric

The prevailing methodology for assessing the accuracy of Large Language Models (LLMs) in multi-choice question

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 02:14:36 UTC from IEEE Xplore. Restrictions apply.

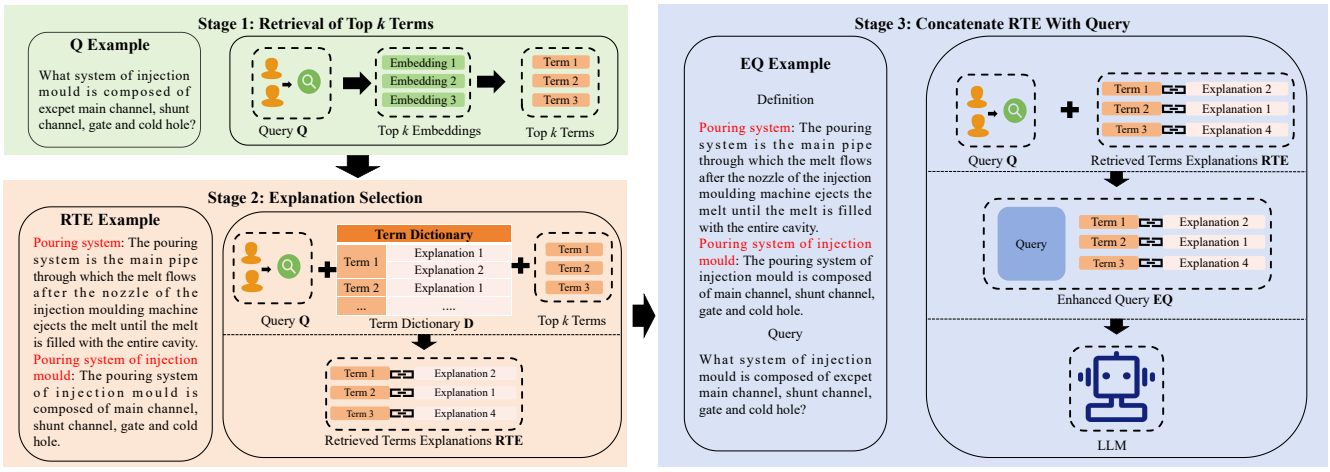


Fig. 4. A Schematic Overview of the Three-Stage Term Explanation Enhancement Process, depicting the integration of term embeddings with a Term Dictionary to facilitate precise retrieval and explanation selection. This methodology culminates in the construction of an Enhanced Query, strategically augmented with contextually relevant explanations for advanced LLM fine-tuning, ensuring domain-specific learning objectives are effectively met.

scenarios predominantly utilizes the Next Token Prediction (NTP) approach. This approach gauges the model's response based on the highest probability token among the provided options A, B, C, D, as cited in [13], [14]. However, this method exhibits limitations in specific domains where LLMs may lack pertinent knowledge, often resulting in outputs like 'fail to judge' or 'no correct answer'. Such responses highlight the inadequacy of the NTP approach in fully capturing the model's capabilities, as illustrated in Fig. 5.

To address these shortcomings, we introduce the Contextualized Adaptive Response Assessment (CARA) method. CARA expands the answer choices by including two additional options: 'E. fail to judge' and 'F. no correct answer'. The accuracy of the model is then evaluated through three distinct sub-methods:

1) **Next Token Prediction (NTP)**: This traditional method selects the token with the highest predictive probability.

$$\text{NTP} = \left[a = \underset{x \in A,B,C,D,E,F}{\text{argmax}} \; P(x|\mathbf{EQ}) \right] \tag{5}$$

Here, a denotes the actual answer.

2) **Bidirectional Matching (BM)**: Utilizes regular expressions to verify the presence of the predicted answer in the model's output and vice versa, thus enhancing the validity of the assessment.

$$\text{BM} = \text{RM}(\text{O}, a) \& \text{RM}(a, \text{O}) \tag{6}$$

In this context, O represents the model's output, and RM performs a check for the existence of the first character of the first string in the second string.

3) **Most Similar Option Matching (MSOM)**: Determines the closest match between the model's output and the available options using vector similarity metrics.

$$\text{MSOM} = \left[a = \underset{x \in \text{Options}}{\text{argmax}} \; \text{SimTF-IDF}(\text{O}, x) \right] \tag{7}$$

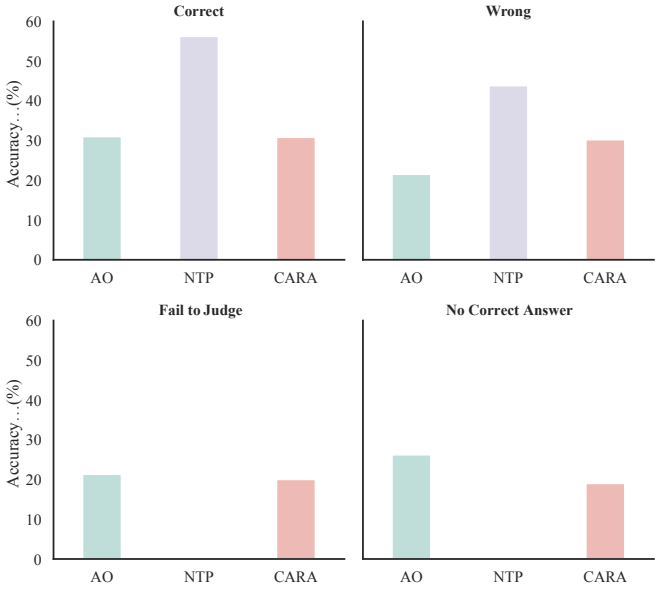


Fig. 5. The performance of original Baichuan2-13b on the validation set, AO represents the actual output of manual evaluation, while NTP represents the Next Token accuracy, it can be seen that there is a huge gap between NTP and the actual output of the model, while CARA can reflect the output of the model closer.

CARA posits that a model's output is deemed accurate if all three sub-methods—NTP, BM, and MSOM—concurrently yield a True result. This multifaceted approach provides a more comprehensive and precise evaluation of a model's performance, especially in domain-specific question-answering tasks where the complexity and nuances of responses necessitate a robust and multifaceted assessment framework.

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 02:14:36 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON OF ACCURACY ACROSS BAICHUAN2-13B MODELS WITH DIFFERENT METRIC METHODS AND GPT.

Method & Model	NTP (%)	BM (%)	MSOM (%)	CARA (%)
Baseline LLM	53.2	48.6	39.0	31.0
LLM Fine-tuned without Retrieval	62.4	60.8	61.2	60.4
LLM Fine-tuned + Sparse Retrievers	66.8	64.6	66.2	64.4
LLM Fine-tuned + Dense Retrievers	67.8	65.8	65.4	65.0
LLM Pretrain + Fine-tuned	62.4	60.8	62.2	60.6
ITEM (ours)	**82.6**	**82.6**	**82.0**	**82.0**
GPT-3.5-turbo	–	53.6	53.6	53.6
GPT-4.0	–	77.8	77.2	77.0

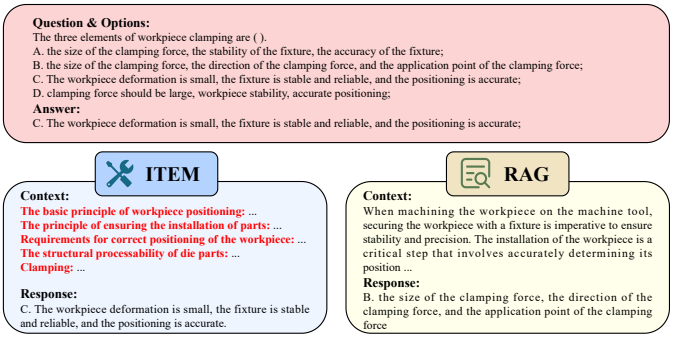


Fig. 6. Case analysis on CMMD validation set. Our ITEM yields the correct answer.

B. Experiment Setup

In our experiment, the CMMD served as the dataset. These were conducted on a robust server configuration: a 48-core Intel(R) Xeon(R) Silver 4310 CPU complemented by 256GB RAM and an A40 GPU with 48GB VRAM. Given the Chinese-centric nature of the dataset, the m3e-base Chinese embedding model was selected for embedding purposes in ITEM. Embeddings were stored in the Chroma vector library. For the choice of LLM, we focused on the Baichuan2 [22]and ChatGLM series [23], selected for their demonstrated efficacy in Chinese language processing. Our training configuration included the application of LoRA [26], where r was set to 64 and α to 16, alongside a fine-tuning stage dropout rate of 0.1. The maximum lengths for input and output were capped at 2048. All of our fine-tuning were 10 epochs. Answer generation was facilitated using a greedy decoding algorithm.

C. Baseline

The model we chose is Baichuan2-13b due to its excellent performance in the Chinese domain. The benchmark includes four configurations:

1) LLM without fine-tuning as the baseline.
2) LLM fine-tuned without retrieval enhancement.
3) LLM fine-tuned and evaluated with the inclusion of sparse retrievers using the BM25 algorithm.
4) LLM fine-tuned and evaluated with the inclusion of dense retrievers using the m3e-base embedding model.
5) LLM pre-trained with the documents, then fine-tuned.

Each configuration is designed to assess the impact of incremental enhancements on the LLM's performance. We limit retrieval token length to 350 and include the top k=5 tokens in our ITEM. Each term with its explanation averages 70 tokens, matching the typical length in retrieval.

A 3-shot assessment was carried out utilizing both GPT-3.5-turbo and the technologically superior GPT-4.0 models to fortify the credibility of our baseline evaluation.

D. Results

Our experimental results, outlined in Table II, demonstrate the effectiveness of the ITEM. The ITEM demonstrated a pronounced superiority over the foundational base model in the research presented, exhibiting significant enhancements in all evaluation methods.

Simultaneously, we also discerned that the gap in accuracy between Next Token Prediction and CARA diminishes following the fine-tuning process. This trend was consistent, irrespective of the integration of retrieval enhancement. This ancillary finding subtly underscores the high caliber of our fine-tuning set. The model becomes more adept in domain-specific knowledge, decreasing the instances of ambiguous or non-committal responses.

The ITEM achieves this within the constrained token span typical of RAG, indicating a strategic balance between token usage and retrieval efficacy. A focal point of our study is the exploration of the optimal number of top k terms, which we intend to further investigate in our ablation studies. Additionally, our study involved a comparative analysis of various methods in incorporating knowledge documents into the LLM inference process. These include pre-training fine-tuning strategies, and using the fine-tuned model to add different methodologies of RAG in the inference stage. Despite these diverse methods, our ITEM method consistently exhibited superior performance. The ITEM surpasses the best-performing baseline model in Next Token Prediction by 14.8% in accuracy and in CARA by 17.0% in accuracy. Our findings suggest that ITEM's innovative method significantly amplifies model performance, validating its potential as a new benchmark in the domain.

The superior performance of ITEM over GPT-4.0 can demonstrate the superiority of our retrieval method, which can outperform its capabilities in specialized domains when the model performance is much weaker than GPT. Our ITEM outperforms GPT-4.0 by 5.0%. This success highlights ITEM's

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 02:14:36 UTC from IEEE Xplore. Restrictions apply.

robustness in contexts where GPT-4.0 faces limitations, underlining the importance of tailored datasets and domain-specific optimization in evaluating and enhancing model performance.

E. Case Study

Given the question and options in Fig. 6, we compare the output of our ITEM and RAG (Dense Retrievers). In this context, the ITEM demonstrates a comprehensive understanding by accurately identifying critical elements of workpiece clamping. It elucidates the necessity of minimal workpiece deformation, fixture stability, and precise positioning, underscoring the intricate balance required for optimal manufacturing outcomes. On the other hand, while competently summarizing the essential steps of workpiece installation and the clamping process, RAG appears to lack the nuanced details that ITEM encompasses. In conclusion, ITEM outshines RAG in delivering a more in-depth, knowledgeable, and elaborate context.

F. Ablation Study

We investigate whether enhanced query participates in the fine-tuning stage and the impact of different top k on our method.

TABLE III
COMPARISON OF RETRIEVAL METHODS UNDER CARA EVALUATION METHOD WITH AND WITHOUT ENHANCED QUERY

Method	Accuracy w/o EQ (%)	Accuracy w/ EQ (%)
Sparse Retrievers	64.4	62.4
Dense Retrievers	65.0	68.2
ITEM	62.4	82.0

1) Fine-tuning with **EQ***:* In the analysis presented in Table III, we observe that the benefits of employing enhanced queries in the fine-tuning phase in RAG are marginal. This observation aligns with our initial hypothesis that simply retrieving context with a high degree of similarity to the query is not necessarily advantageous for the model's output. Additionally, our study indicates that fine-tuning without **EQ** in ITEM results in limited performance gains due to the fact that the model has not learned the structure of the **EQ**) inputs, after fine-tuning training the model mastered the structure of **EQ** and demonstrated strong capabilities.

2) Impact of the number of iterations k: In our ITEM, we have the flexibility to select the top k terms for enhancement. This section explores the impact of varying the number of k on the ITEM's effectiveness. In Fig. 7, we observe that the optimal number of k for achieving the best accuracy differs across models of varying sizes. For the ChatGLM3-6b and Baichuan2-7b models, setting k to 4 yields the best results, whereas for the larger Baichuan2-13b model, the optimal performance is achieved when k is set to 5. This variation can be attributed to the phenomenon of Context Truncation, which occurs when the model input becomes excessively long, leading to a loss of contextual information and thereby impacting the accuracy and relevance of the generated responses [31]. It is worth mentioning that our ITEM achieved a accuracy of 69.2% when k is set to 1, surpassing the best performance of RAG with 350 tokens, which is 65.0%. This represents an 80% token reduction.

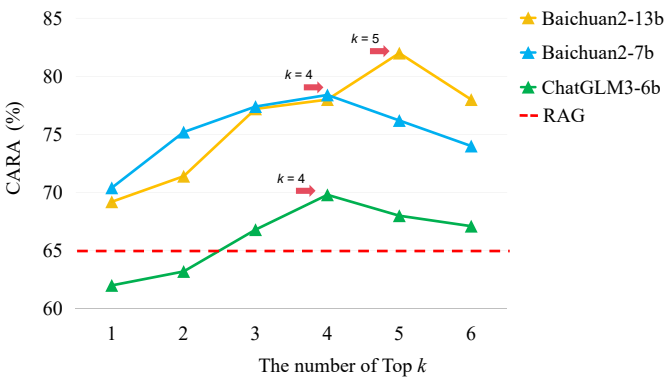


Fig. 7. Performance of different models with different numbers of top k.

VI. CONCLUSION AND FUTURE WORK

In our paper, we present the *Integrated Term Enhancement Methodology* (ITEM), proposing Chinese Mould Manufacturing Dataset (CMMD) and Contextualized Adaptive Response Assessment (CARA) accuracy evaluation method. ITEM's central feature, the AI-assisted Term Dictionary, is specifically designed to bridge complex domain terms with their explanation facilitating a deeper understanding of domain-specific knowledge. ITEM outperforms the RAG method in accuracy by 17.0% and requires only 80% of its token count to achieve the same accuracy. Additionally, ITEM surpasses GPT-4 by 5.0% in accuracy, offering a new perspective on improving LLM outputs by retrieving more relevant context for queries.

Although our ITEM is highly efficient, it relies on GPT to construct the Term Dictionary. In future research, we aim to explore whether knowledge-distilled models could replace GPT's role in this process, reducing reliance on the GPT model and potentially decreasing associated costs. This exploration aims to broaden the potential use cases for ITEM, making it more versatile and accessible for various applications.

ACKNOWLEDGMENT

This work was supported by the National Key Research and Development Program of China (Grant No. 2022YFF0902003).

REFERENCES

[1] J. Ye, X. Chen, N. Xu, C. Zu, Z. Shao, S. Liu, Y. Cui, Z. Zhou, C. Gong, Y. Shen, et al., "A comprehensive capability analysis of gpt-3 and gpt-3.5 series models," arXiv preprint arXiv:2303.10420, 2023.

[2] OpenAI, J. Achiam, S. Adler, S. Agarwal, L. Ahmad, I. Akkaya, F. L. Aleman, D. Almeida, J. Altenschmidt, S. Altman, et al., "GPT-4 Technical Report," 2023, arXiv:2303.08774 [cs.CL].

[3] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "React: Synergizing reasoning and acting in language models," arXiv preprint arXiv:2210.03629, 2022.

[4] Y. Bang, S. Cahyawijaya, N. Lee, W. Dai, D. Su, B. Wilie, H. Lovenia, Z. Ji, T. Yu, W. Chung, et al., "A multitask, multilingual, multimodal evaluation of chatgpt on reasoning, hallucination, and interactivity," arXiv preprint arXiv:2302.04023, 2023.

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 02:14:36 UTC from IEEE Xplore. Restrictions apply.

[5] M. Sallam, "ChatGPT utility in healthcare education, research, and practice: Systematic review on the promising perspectives and valid concerns," in Healthcare, vol. 11, no. 6, pp. 887, 2023, MDPI.
[6] Z. Kraljevic, A. Shek, D. Bean, R. Bendayan, J. Teo, and R. Dobson, "MedGPT: Medical concept prediction from clinical narratives," arXiv preprint arXiv:2107.03134, 2021.
[7] R. Luo, L. Sun, Y. Xia, T. Qin, S. Zhang, H. Poon, and T.-Y. Liu, "BioGPT: Generative pre-trained transformer for biomedical text generation and mining," Briefings in Bioinformatics, vol. 23, no. 6, pp. bbac409, 2022, Oxford University Press.
[8] Z. Bi, N. Zhang, Y. Xue, Y. Ou, D. Ji, G. Zheng, and H. Chen, "Oceangpt: A large language model for ocean science tasks," arXiv preprint arXiv:2310.02031, 2023.
[9] G. Singh and A. Verma, "A brief review on injection moulding manufacturing process," Materials Today: Proceedings, vol. 4, no. 2, pp. 1423–1433, 2017, Elsevier.
[10] R. Rai, M. K. Tiwari, D. Ivanov, and A. Dolgui, "Machine learning in manufacturing and industry 4.0 applications," International Journal of Production Research, vol. 59, no. 16, pp. 4773–4778, 2021, Taylor & Francis.
[11] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al., "Language models are few-shot learners," Advances in Neural Information Processing Systems, vol. 33, pp. 1877–1901, 2020.
[12] Kurt Shuster, Spencer Poff, Moya Chen, Douwe Kiela, and Jason Weston, "Retrieval augmentation reduces hallucination in conversation," arXiv preprint arXiv:2104.07567, 2021.
[13] D. Hendrycks, C. Burns, S. Basart, A. Zou, M. Mazeika, D. Song, and J. Steinhardt, "Measuring massive multitask language understanding," in Proceedings of the International Conference on Learning Representations (ICLR), 2021.
[14] Y. Huang, Y. Bai, Z. Zhu, J. Zhang, J. Zhang, T. Su, J. Liu, C. Lv, Y. Zhang, J. Lei, Y. Fu, M. Sun, and J. He, "C-Eval: A multi-level multi-discipline Chinese evaluation suite for foundation models," in Advances in Neural Information Processing Systems, 2023.
[15] R. Anil, A. M. Dai, O. Firat, M. Johnson, D. Lepikhin, A. Passos, S. Shakeri, E. Taropa, P. Bailey, Z. Chen, et al., "Palm 2 technical report," arXiv preprint arXiv:2305.10403, 2023.
[16] S. Zhang, S. Roller, N. Goyal, M. Artetxe, M. Chen, S. Chen, C. Dewan, M. Diab, X. Li, X. V. Lin, et al., "OPT: Open pre-trained transformer language models," arXiv preprint arXiv:2205.01068, 2022.
[17] D. Chen, A. Fisch, J. Weston, and A. Bordes, "Reading wikipedia to answer open-domain questions," arXiv preprint arXiv:1704.00051, 2017.
[18] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 6769–6781.
[19] J. Ramos, et al., "Using tf-idf to determine word relevance in document queries," in Proceedings of the first instructional conference on machine learning, vol. 242, no. 1, pp. 29–48, 2003, Citeseer.
[20] S. Robertson and H. Zaragoza, et al., "The probabilistic relevance framework: BM25 and beyond," Foundations and Trends® in Information Retrieval, vol. 3, no. 4, pp. 333–389, 2009, Now Publishers, Inc.
[21] Y. Wang, Q. Sun, and S. He, "M3E: Moka Massive Mixed Embedding Model," 2023.
[22] A. Yang, B. Xiao, B. Wang, B. Zhang, C. Bian, C. Yin, C. Lv, D. Pan, D. Wang, D. Yan, et al., "Baichuan 2: Open large-scale language models," arXiv preprint arXiv:2309.10305, 2023.
[23] Z. Du, Y. Qian, X. Liu, M. Ding, J. Qiu, Z. Yang, and J. Tang, "GLM: General language model pretraining with autoregressive blank infilling," in Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 320–335, 2022.
[24] U. Frank, "Domain-specific modeling languages: Requirements analysis and design guidelines," in Domain Engineering: Product Lines, Languages, and Conceptual Models, Springer, 2013, pp. 133–157.
[25] J. Wei, Y. Tay, R. Bommasani, C. Raffel, B. Zoph, S. Borgeaud, D. Yogatama, M. Bosma, D. Zhou, D. Metzler, et al., "Emergent abilities of large language models," arXiv preprint arXiv:2206.07682, 2022.
[26] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," arXiv preprint arXiv:2106.09685, 2021.
[27] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, "Large language models in medicine," Nature Medicine, vol. 29, no. 8, pp. 1930–1940, 2023, Nature Publishing Group US, New York.
[28] W. Zhang, H. Shen, T. Lei, Q. Wang, D. Peng, and X. Wang, "GLQA: A generation-based method for legal question answering," in 2023 International Joint Conference on Neural Networks (IJCNN), IEEE, 2023, pp. 1–8.
[29] J. Liu, L. Liangzhi, T. Xiang, B. Wang, and Y. Qian, "TCRA-LLM: Token compression retrieval augmented large language model for inference cost reduction," arXiv preprint arXiv:2310.15556, 2023.
[30] X. Ma, Y. Gong, P. He, H. Zhao, and N. Duan, "Query rewriting for retrieval-augmented large language models," arXiv preprint arXiv:2305.14283, 2023.
[31] Y. Li, B. Dong, C. Lin, and F. Guerin, "Compressing context to enhance inference efficiency of large language models," arXiv preprint arXiv:2310.06201, 2023.
[32] X. Chen, K. Lakhotia, B. Oğuz, A. Gupta, P. Lewis, S. Peshterliev, Y. Mehdad, S. Gupta, and W.-t. Yih, "Salient phrase aware dense retrieval: Can a dense retriever imitate a sparse one?," arXiv preprint arXiv:2110.06918, 2021.
[33] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," ACM Comput. Surv., vol. 55, no. 12, 2023.
[34] C.-C. Hsu, E. Lind, L. Soldaini, and A. Moschitti, "Answer generation for retrieval-based question answering systems," in Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021, 2021, pp. 4276–4282.
[35] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," arXiv preprint arXiv:2307.03172, 2023.

Authorized licensed use limited to: University of Kansas Libraries. Downloaded on October 13,2026 at 02:14:36 UTC from IEEE Xplore. Restrictions apply.